THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

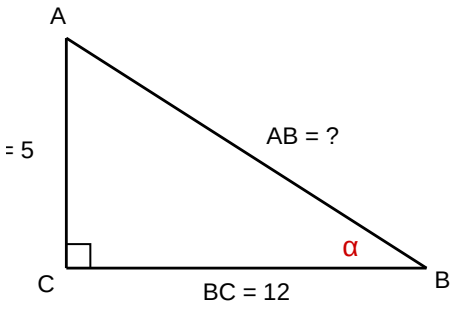
ANSWER KEY & MARKING SCHEME — MATHEMATICS · CLASS 10 · SET 02

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION – A (Multiple Choice Questions · 20 × 1 = 20) **20 questions · 20 marks**

Q1–Q18 are multiple-choice questions. Q19–Q20: a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

1. The value of cos α from the given figure is



(a) $\frac{12}{13}$ (b) $\frac{5}{13}$ (c) $\frac{5}{12}$ (d) $\frac{13}{12}$

Ans: (a) AB = √(12² + 5²) = 13; cos α = BC/AB = 12/13.

2. In ΔABC and ΔDEF, AB/DE = BC/EF and ∠B = ∠E. Then

(a) ΔABC ~ ΔEDF (b) ΔABC ~ ΔDEF (c) ΔABC ~ ΔFED (d) ΔCBA ~ ΔDEF

Ans: (b) Two sides proportional and the included angles equal ⇒ SAS similarity with A↔D, B↔E, C↔F.

3. The HCF and LCM of 15, 20 and 25 are respectively

(a) 5, 150 (b) 1, 300 (c) 5, 300 (d) 10, 300

Ans: (c) 15 = 3·5, 20 = 2²·5, 25 = 5². HCF = 5; LCM = 2²·3·5² = 300.

4. Two friends were born in a leap year. The probability that they have the same birthday is

(a) $\frac{1}{365}$ (b) $\frac{1}{12}$ (c) $\frac{2}{366}$ (d) $\frac{1}{366}$

Ans: (d) A leap year has 366 days; the second friend's birthday matches the first in 1 of 366 ways.

5. The distance of a point P from the *x*-axis is 5 units. Which of the following cannot be the coordinates of P?

(a) (3, 5) (b) (5, 3) (c) (−3, −5) (d) (0, 5)

Ans: (b) Distance from x-axis = |y| = 5; (5, 3) has |y| = 3.

6. The value of (4 + √3)(4 − √3) − (√5)² is

(a) 11 (b) 13 (c) 8 (d) 18

Ans: (c) (16 − 3) − 5 = 8.

7. √8 + √18 − √50 equals

(a) 0 (b) 2√2 (c) √26 (d) 10√2

Ans: (a) 2√2 + 3√2 − 5√2 = 0.

8. A taxi charges ₹ 40 for the first kilometre and ₹ 12 for each additional kilometre. The fare for a 10 km journey is

(a) ₹ 160 (b) ₹ 120 (c) ₹ 172 (d) ₹ 148

Ans: (d) Fares 40, 52, 64, … form an AP; $a_{10} = 40 + 9 \times 12 = 148$.

9. The pair of equations $kx + 2y = 7$ and $3x + y = 5$ has no solution when k equals

(a) 2 (b) 3 (c) 6 (d) 7

Ans: (c) $k/3 = 2/1 \Rightarrow k = 6$; then $2/1 \neq 7/5$, so the lines are parallel.

10. The expression $(1 - \cos^2\theta)\,\text{cosec}^2\theta$ equals

(a) 0 (b) 1 (c) $\sin^2\theta$ (d) $\text{cosec}^2\theta$

Ans: (b) $(1 - \cos^2\theta) = \sin^2\theta$; $\sin^2\theta \cdot \text{cosec}^2\theta = 1$.

11. The mid-point of the line segment joining A$(2a, 4)$ and B$(-2, 2b)$ is $(1, 2a + 1)$. Then $a + b$ equals

(a) 5 (b) 4 (c) 6 (d) 3

Ans: (a) $(2a - 2)/2 = 1 \Rightarrow a = 2$; $(4 + 2b)/2 = 2a + 1 = 5 \Rightarrow b = 3$; $a + b = 5$.

12. A 2-digit number less than 60 is selected at random. The probability that it is a multiple of 4 and 6 but not of 5 is

(a) $\frac{1}{10}$ (b) $\frac{1}{12}$ (c) $\frac{4}{59}$ (d) $\frac{2}{25}$

Ans: (d) Numbers 10–59: 50. Multiples of 12: 12, 24, 36, 48 (none divisible by 5) $\Rightarrow 4/50 = 2/25$.

13. The solution of the pair $3x + 2y = 13$ and $3x - 2y = 5$ is

(a) $x = 2, y = 3$ (b) $x = 4, y = 1$ (c) $x = 3, y = 2$ (d) $x = 1, y = 5$

Ans: (c) Adding: $6x = 18 \Rightarrow x = 3$; then $y = 2$.

14. If x and y are different primes, the LCM of x^2y^3 and x^3y^2 is

(a) x^3y^3 (b) x^2y^2 (c) x^5y^5 (d) x^3y^2

Ans: (a) LCM takes the greatest power of each prime.

15. Two dice are thrown together. The probability that the difference of the numbers is 2 is

(a) $\frac{1}{6}$ (b) $\frac{2}{9}$ (c) $\frac{5}{18}$ (d) $\frac{1}{9}$

Ans: (b) Pairs (1,3), (2,4), (3,5), (4,6) and their reverses: 8 favourable out of 36 = 2/9.

16. If $\sin\theta = \frac{7}{25}$, then $\tan\theta$ equals

(a) $\frac{24}{7}$ (b) $\frac{7}{18}$ (c) $\frac{25}{24}$ (d) $\frac{7}{24}$

Ans: (d) $\cos\theta = 24/25 \Rightarrow \tan\theta = 7/24$.

17. The pair of linear equations $x + 2y = 4$ and $3x + 6y = 12$ represents two lines which are

(a) parallel (b) coincident
(c) intersecting at one point (d) intersecting at two points

Ans: (b) $1/3 = 2/6 = 4/12 \Rightarrow$ coincident lines.

18. The common difference of the AP $\frac{1}{p}, \frac{1-p}{p}, \frac{1-2p}{p}, \ldots$ is

(a) 1 (b) p (c) −p (d) −1

Ans: (d) $d = (1 - p)/p - 1/p = -p/p = -1$.

19. **Assertion (A):** The distance between the points (0, 0) and (−3, 4) is 5 units.
Reason (R): The distance between (x_1, y_1) and (x_2, y_2) is $\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$.

Ans: (a) $\sqrt{(9 + 16)} = 5$ using the distance formula.

20. **Assertion (A):** For every angle θ, $(\sin\theta + \cos\theta)^2 = 1 + 2\sin\theta\cos\theta$.
Reason (R): $\sin\theta + \cos\theta = 1$ for every angle θ.

> **Ans: (c)** A is true by expanding; R is false (e.g. θ = 45°).

SECTION – B (Very Short Answer Questions · 5 × 2 = 10) — 5 questions · 10 marks

21. Use Euclid's division algorithm to find the HCF of 96 and 404.

> 404 = 96×4 + 20; 96 = 20×4 + 16 (1); 20 = 16×1 + 4; 16 = 4×4 + 0 ⇒ HCF = 4 (1).

O R

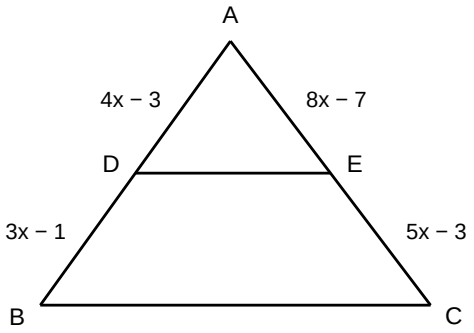
Two traffic lights change after every 40 s and 60 s. If they change together at 8:00:00 a.m., when will they next change together?

> LCM(40, 60) = 120 s (1) = 2 minutes ⇒ 8:02:00 a.m. (1).

22. A girl 1.5 m tall casts a shadow 2 m long. At the same time a tree casts a shadow 12 m long. Find the height of the tree.

> Similar right triangles (1): 1.5/2 = h/12 ⇒ h = 9 m (1).

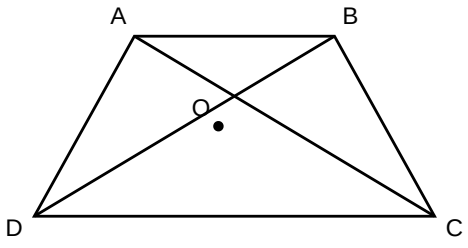
23. In ΔABC, DE ∥ BC with AD = 4*x* − 3, DB = 3*x* − 1, AE = 8*x* − 7 and EC = 5*x* − 3. Find *x*.



> BPT: (4x − 3)(5x − 3) = (3x − 1)(8x − 7) ⇒ $2x^2 - x - 1 = 0$ (1) ⇒ x = 1 (x = −½ rejected as lengths would be negative) (1).

O R

In the trapezium ABCD (AB ∥ DC), the diagonals meet at O. Prove that OA/OC = OB/OD.



> In ΔAOB and ΔCOD: ∠OAB = ∠OCD, ∠OBA = ∠ODC (alternate angles) ⇒ ΔAOB ~ ΔCOD (1) ⇒ OA/OC = OB/OD (1).

24. A ladder 10 m long just reaches a window 5√3 m above the ground. Find the angle the ladder makes with the ground.

> $\sin\theta = 5\sqrt{3}/10 = \sqrt{3}/2$ (1) ⇒ θ = 60° (1).

25. The LCM of two numbers is 2400 and their HCF is 20. If one number is 300, find the other.

> Product = HCF × LCM = 20 × 2400 = 48000 (1) ⇒ other = 48000/300 = 160 (1).

SECTION – C (Short Answer Questions · 6 × 3 = 18) — 6 questions · 18 marks

26. Prove that $5 + 3\sqrt{2}$ is irrational, given that $\sqrt{2}$ is irrational.

> Suppose $5 + 3\sqrt{2} = p/q$, $q \neq 0$, p, q integers (1) ⇒ $\sqrt{2} = (p - 5q)/(3q)$, which is rational (1). This contradicts the irrationality of $\sqrt{2}$, so $5 + 3\sqrt{2}$ is irrational (1).

27. Solve for x and y: $7x - 3y = 31$; $9x - 5y = 41$.

> Multiply the first by 5 and the second by 3: 35x − 15y = 155; 27x − 15y = 123 (1). Subtract: 8x = 32 ⇒ x = 4 (1). Then y = (28 − 31)/3 = −1 (1).

28. Find the sum of all two-digit numbers divisible by 7.

> 14, 21, …, 98: a = 14, d = 7, l = 98 ⇒ n = 13 (1½). S = 13/2 (14 + 98) = 728 (1½).

OR

The sum of the first n terms of an AP is $3n^2 + 5n$. Find its 20th term.

> $S_{20} = 3(400) + 100 = 1300$ (1); $S_{19} = 3(361) + 95 = 1178$ (1); $a_{20} = 1300 - 1178 = 122$ (1).

29. Find the point on the x-axis which is equidistant from (2, −5) and (−2, 9).

> Let (x, 0): $(x - 2)^2 + 25 = (x + 2)^2 + 81$ (1½) ⇒ −8x = 56 ⇒ x = −7. Point (−7, 0) (1½).

OR

Find a relation between x and y such that the point P(x, y) is equidistant from A(5, 1) and B(−1, 5).

> $PA^2 = PB^2$ ⇒ $(x - 5)^2 + (y - 1)^2 = (x + 1)^2 + (y - 5)^2$ (1½) ⇒ −12x + 8y = 0 ⇒ 3x = 2y (1½).

30. If $7 \sin^2 \theta + 3 \cos^2 \theta = 4$, show that $\tan \theta = 1/\sqrt{3}$.

> $7 \sin^2\theta + 3(1 - \sin^2\theta) = 4 \Rightarrow 4 \sin^2\theta = 1$ (1) ⇒ $\sin \theta = ½ \Rightarrow \theta = 30°$ (1) ⇒ $\tan \theta = 1/\sqrt{3}$ (1).

31. Three coins are tossed together. Find the probability of getting (a) exactly two tails (b) at least two heads (c) no head.

> Sample space of 8 outcomes (½). (a) HTT, THT, TTH ⇒ 3/8 (1). (b) HHH, HHT, HTH, THH ⇒ 4/8 = 1/2 (1). (c) TTT ⇒ 1/8 (½).

SECTION – D (Long Answer Questions · 4 × 5 = 20) — 4 questions · 20 marks

32. State and prove the Pythagoras theorem.

> Statement: in a right triangle the square on the hypotenuse equals the sum of the squares on the other two sides (1). Given ΔABC right-angled at B; draw BD ⊥ AC (1). ΔADB ~ ΔABC ⇒ $AB^2 = AD \cdot AC$; ΔBDC ~ ΔABC ⇒ $BC^2 = DC \cdot AC$ (2). Adding: $AB^2 + BC^2 = AC(AD + DC) = AC^2$ (1).

33. A boat goes 30 km upstream and 44 km downstream in 10 hours; it goes 40 km upstream and 55 km downstream in 13 hours. Find the speed of the stream and the speed of the boat in still water.

> Let upstream speed u, downstream d: 30/u + 44/d = 10; 40/u + 55/d = 13 (1). Put p = 1/u, q = 1/d: 30p + 44q = 10, 40p + 55q = 13 (1). Solve: q = 1/11, p = 1/5 (2) ⇒ u = 5, d = 11. Boat = (11 + 5)/2 = 8 km/h; stream = (11 − 5)/2 = 3 km/h (1).

OR

Draw the graphs of $x + y = 5$ and $2x - y = 1$ on the same axes. Find their point of intersection and the points where they meet the y-axis.

> Tables: x + y = 5: (0, 5), (2, 3), (5, 0); 2x − y = 1: (0, −1), (1, 1), (2, 3) (2). Graph (1½). Intersection (2, 3) (1). y-intercepts (0, 5) and (0, −1) (½).

34. From the top of a 60 m high building, the angles of depression of the top and the bottom of a tower are 30° and 60° respectively. Find the height of the tower.

Let building AB = 60 m, tower CD. Horizontal distance BD = 60/tan 60° = 20√3 m (2). Drop to top of tower: BD tan 30° = 20√3 × 1/√3 = 20 m (2). Height of tower = 60 − 20 = 40 m (1).

35. Prove that $\frac{\tan\theta + \sec\theta - 1}{\tan\theta - \sec\theta + 1} = \frac{1 + \sin\theta}{\cos\theta}$.

Use $1 = \sec^2\theta - \tan^2\theta$: numerator = $(\sec\theta + \tan\theta) - (\sec\theta - \tan\theta)(\sec\theta + \tan\theta) = (\sec\theta + \tan\theta)(1 - \sec\theta + \tan\theta)$ (2½). So LHS = $\sec\theta + \tan\theta$ (1) = $(1 + \sin\theta)/\cos\theta$ (1½).

OR

If $\tan(A + B) = \sqrt{3}$ and $\tan(A - B) = 1/\sqrt{3}$, $0° < A + B \le 90°$, $A > B$, find A and B. Hence evaluate $\sin(A + B) + \cos(A - B)$.

$A + B = 60°$, $A - B = 30° \Rightarrow A = 45°$, $B = 15°$ (3). $\sin 60° + \cos 30° = \sqrt{3}/2 + \sqrt{3}/2 = \sqrt{3}$ (2).

SECTION – E (Case Study Based Questions · 3 × 4 = 12) — 3 questions · 12 marks

36. At the sports-day lucky draw, 25 chits numbered 1 to 25 are placed in a box. A student draws one chit at random. Multiples of 4 win a badge, prime numbers win a pen, and a perfect square or a multiple of 10 wins a bonus ticket.

(i) Find the probability of winning a badge.

4, 8, 12, 16, 20, 24 ⇒ 6/25 (1).

(ii) Find the probability that the chit drawn shows a prime number.

2, 3, 5, 7, 11, 13, 17, 19, 23 ⇒ 9/25 (1).

(iii) Find the probability of winning a bonus ticket.

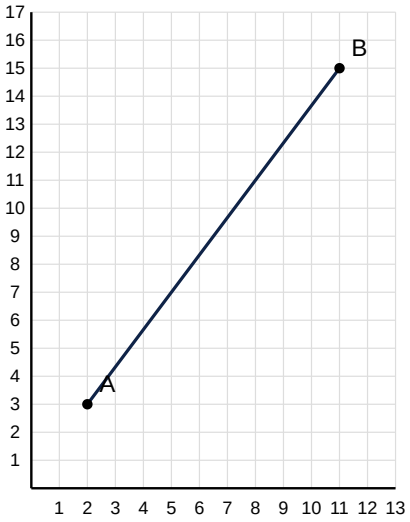
Squares: 1, 4, 9, 16, 25 (5); multiples of 10: 10, 20 (2); no overlap ⇒ 7/25 (2).

OR

(iii) The first student draws chit 9 and keeps it. Find the probability that the next student draws a prime number.

24 chits remain; 9 primes ⇒ 9/24 = 3/8 (2).

37. A treasure-hunt map of a school ground is drawn on a coordinate grid (1 unit = 10 m). The flag is at A(2, 3) and the treasure box is at B(11, 15). Teams must walk straight from A to B.



(i) Find the distance AB in grid units.

AB = $\sqrt{(9^2 + 12^2)}$ = 15 units (1).

(ii) Find the coordinates of the midpoint of AB.

> (6.5, 9) (1).

(iii) A clue is hidden at point P on AB such that AP : PB = 1 : 2. Find the coordinates of P.

> P = ((1×11 + 2×2)/3, (1×15 + 2×3)/3) = (5, 7) (2).

OR

(iii) Find the point on the *x*-axis which is equidistant from A and B.

> $(x - 2)^2 + 9 = (x - 11)^2 + 225 \Rightarrow 18x = 333 \Rightarrow x = 37/2$; point (37/2, 0) (2).

38. Tarun is training for a 10 km run. On the first day he jogs 800 m and every day he increases the distance by 150 m. The distances covered on successive days form an AP.

(i) Write the first term and the common difference.

> a = 800 m, d = 150 m (1).

(ii) Write the expression for the distance he jogs on the *n*th day.

> $a_n = 800 + (n - 1) \times 150 = 650 + 150n$ (1).

(iii) How many metres does he jog on the 8th day?

> $a_8 = 800 + 7 \times 150 = 1850$ m (2).

OR

(iii) What is the total distance he covers in the first 10 days?

> $S_{10} = 10/2\ [1600 + 9 \times 150] = 5 \times 2950 = 14750$ m (2).